

BREATHBOX · DESIGN DECISION RECORD

BreatheBox TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	BBX-DDR-001	Version	0.3
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breathebox		
Source	docs/decisions/0001-trl2-review-decisions.md @ d61ee78		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-26	Amish Chadha	O3 decided by Amish, 2026-09-26: budget \$255 (BBX-DDR-002)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 recommendations adopted for TRL 3 under Amish's 2026-09-25 instruction, open for his review, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** items A1 to A6 and O2 decided by Amish, 2026-09-25: go with recommendation (see BBX-DDR-002); item O1 remains proposed, awaiting Amish; O3 was decided by Amish on 2026-09-26 (budget \$255, BBX-DDR-002).

■ Context

The TRL 2 review note (docs/REVIEW.md , session of 2026-09-25) listed seven items as "Proposed, awaiting Amish", six of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He did not review this repo's items one by one. The recommendations are therefore adopted as the basis for the TRL 3 calculations, model and drawing, and stay open for his review. Nothing in version 0.1 of this record was decided by Amish. On 2026-09-25 he accepted all recommendations ("i accept all your recommendations, go with them across all repos"); the status column now shows this, and BBX-DDR-002 lists the changes.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 session) and BBX-PRC-001. They are not repeated here.

■ Decision

Table 1. Items adopted for TRL 3, since decided by Amish (BBX-DDR-002).

#	Item	Status	Where it now lives
A1	Budget: recommendation (a), raise <code>budget_usd</code> from \$220 to \$250 and keep the bought counterflow core and the CO2 sensor	Decided by Amish, 2026-09-25: go with recommendation. <code>budget_usd</code> is now \$250 (BBX-DDR-002); the priced BOM is \$255, \$5 over	BBX-CAL-001 section 9, BBX-REQ-001 R15, <code>bom/bom.csv</code>
A2	Window type for the first version: vertical sliding sash	Decided by Amish, 2026-09-25: go with recommendation. Casement and tilt-and-turn inserts are a later variant	BBX-PRC-001 v0.3, BBX-REQ-001 R8, <code>cad/src/model.py</code>
A3	Core type: bought polymer counterflow plate core (sensible only)	Decided by Amish, 2026-09-25: go with recommendation	BBX-PRC-001 v0.3, BBX-CAL-001 section 2, <code>bom/bom.csv</code> line 2
A4	Frost strategy: slow the supply fan, with the combustion-appliance warning	Decided by Amish, 2026-09-25: go with recommendation	BBX-PRC-001 v0.3 (Safety), BBX-CAL-001 section 6
A5	Nominal airflow: 50 m ³ /h with CO2-driven boost to 70 m ³ /h	Decided by Amish, 2026-09-25: go with recommendation	BBX-REQ-001 R1, BBX-CAL-001 sections 3 and 4
A6	Controller: single ESP32-C3 class module with an SCD41 class sensor, with sensor checks on CalRig	Decided by Amish, 2026-09-25: go with recommendation	BBX-PRC-001 v0.3, <code>bom/bom.csv</code> line 7

The TRL 2 review made no recommendation to change the pitch or problem wording, so `project.yaml` and `README.md` keep them.

Items that remain open

Table 2. Open items.

#	Item	Status
O1	First co-design partner and region (for example a tenants' group or social landlord in the UK or Canada)	Proposed, awaiting Amish. No recommendation was made at TRL 2
O2	R6 noise shortfall found at TRL 3 (about 39 dB(A) at 50 m ³ /h against 30 dB(A))	Decided by Amish, 2026-09-25: go with recommendation (quiet night mode, plus larger fans or a silencer checked against a real fan datasheet before any change to R6). See BBX-DDR-002
O3	Budget figure: \$255 priced against \$250 in <code>project.yaml</code> (was \$220)	Decided by Amish, 2026-09-26: budget set to \$255 to cover the priced BOM (BBX-DDR-002)

Consequences

- The TRL 3 work is costed and sized for a sash-window unit with a bought counterflow core, two 24 V blowers, a CO2 sensor and a slowed-fan frost mode.
- BBX-CAL-001 shows R6 and R15 not met; neither is resolved by these adoptions.

- The CO2 sensor check depends on CalRig. CalRig's R9 (CO2 span 400 to 2,000 ppm against a reference of $\pm(30 \text{ ppm} + 3 \%)$) covers the 1,000 ppm range BreatheBox needs, and an SCD41 class breakout fits CalRig's 90 x 70 x 50 mm bay (R10). This is consistent with `calrig/docs/REVIEW.md`.
- Amish may reverse any item; the calculations would then be rerun.